

Neon lamp

Y13 (2013) — English translation

The electrical circuit (Fig. 1) consists of an ideal inductor L capacitor C , a battery with EMF U_0 and a nonlinear NE element, the current-voltage characteristic of which is shown in Fig. 2 (gas-filled lamps have a similar characteristic). It is known that the hysteresis zone $\Delta U \ll U_{\text{э}}$; in addition current $I_0 \gg U_0 \sqrt{\frac{C}{L}}$

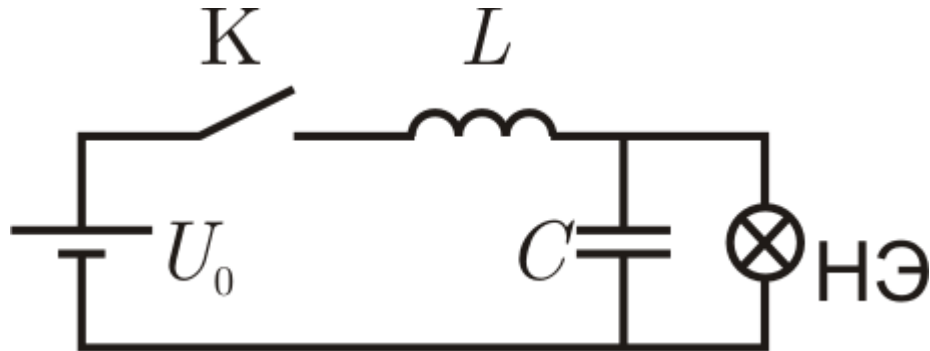


Fig. 1.

Rice. 2. Current-voltage characteristic of a nonlinear element

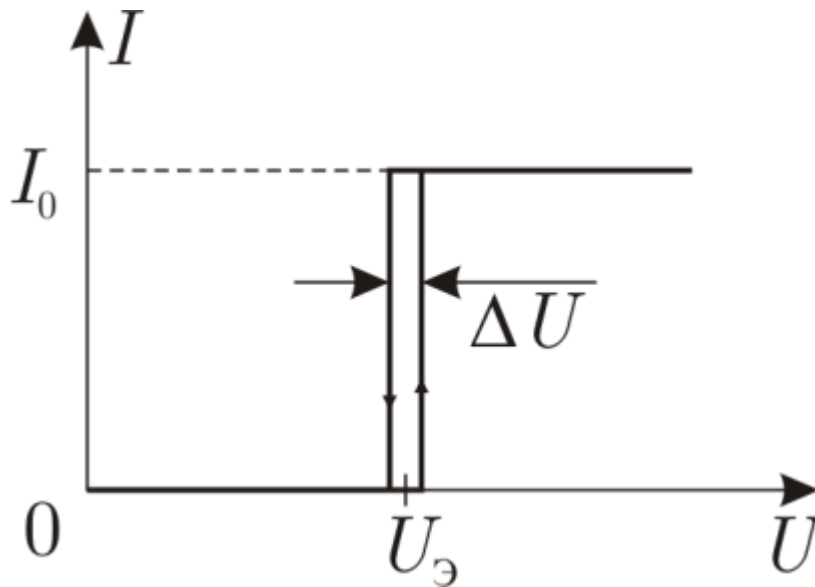


Fig. 2.

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| A1 | Determine how many times the lamp will flash after closing the key K , if $U_0 < U_{\text{э}} < 2U_0$. Draw (qualitatively) the process on the graph $U_C(t)$. | 0 |
| A2 | Determine the steady-state flash frequency f and the average power W dissipated in the lamp if $U_0 = U_{\text{э}}$ | 0 |